

Chuyao Fu

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EDUCATION

Southern University of Science and Technology (SUSTech)

B.Eng. in Electronic Information Engineering

Sep. 2023 – Jun. 2027

GPA: 3.90/4.00 Rank: 3/46

- **Selected Coursework:** Data Structures and Algorithm Analysis, Probability and Statistics, Machine Learning, Robotic Motion and Control, Digital Signal Processing, Signals and Systems, Analog Circuits, Digital Circuits

University of California, Berkeley

Visiting Student

Aug. 2026 – Dec. 2026

PUBLICATIONS

ProDrive: Proactive Planning for Autonomous Driving via Ego-Environment Co-Evolution

Chuyao Fu, Shengzhe Gan, Zhuoli Ouyang, Yuhao Rui, Xiaowei Chi, Sirui Han, Jiankun Wang, Hong Zhang

CVPR 2026 GigaBrain Challenge Workshop [Paper]

Learning Action-Video Compatibility as a Reference-Free Verifier for Embodied World Models

Chuyao Fu, Xiaojie Zhang, Xiaowei Chi, Rongyu Zhang, Yuhao Rui, Jiajun Cao, Yike Guo, Sirui Han, Shanghang Zhang

Under review [Paper]

Token-World: Compact VLM Representations for Policy-Oriented World Modeling

Chuyao Fu, Xiaowei Chi, Haoran Li, Yuhao Rui, Zezhong Qian, Yu-Kai Wang, Xiaojie Zhang, Yunfan Lou, Hongyang Cheng, Yike Guo, Sirui Han, Shanghang Zhang

Submitted to ICRA 2027 [Paper]

AxisField-GS: Feed-Forward Articulation Recovery from Single-State Semantic Gaussian Assets

Shengzhe Gan*, Chuyao Fu*, Yugong Wang, Guangcheng Chen, Hong Zhang. *Equal contribution

Under review [Paper]

Mask World Model: Predicting What Matters for Robust Robot Policy Learning

Yunfan Lou, Xiaowei Chi, Xiaojie Zhang, Zezhong Qian, Chengxuan Li, Rongyu Zhang, Yaoyu Lyu, Guoyu Song, Chuyao Fu, Haoxuan Xu, Pengwei Wang, Shanghang Zhang

ICML 2026 [Paper]

FORCE: Efficient VLA Reinforcement Fine-Tuning via Value-Calibrated Warm-up and Self-Distillation

Shuyi Zhang, Yunfan Lou, Hongyang Cheng, Yichen Guo, Chuyao Fu, Yaoyu Lyu, Xiaojie Zhang, Haoran Li, Pengwei Wang, Zhongyuan Wang, Shanghang Zhang

arXiv preprint, 2026 [Paper]

RESEARCH EXPERIENCE

ProDrive

Robotics and Computer Vision Lab, SUSTech

Autonomous Driving | World-Model-Guided Planning

Supervisors: Prof. Max Q.-H. Meng and Prof. Hong Zhang

- Implemented parallel candidate rollout evaluation and end-to-end planner-world-model training to jointly optimize planning and prediction.
- Achieved 86.6 PDMS on NAVSIM v1, outperforming World4Drive, a closely related world-model reranking baseline, by 1.5 points; accepted to the CVPR 2026 GigaBrain Challenge Workshop.

AxisField-GS

Robotics and Computer Vision Lab, SUSTech

3D Gaussian Splatting | Articulated Digital Twin

Supervisors: Prof. Max Q.-H. Meng and Prof. Hong Zhang

- Implemented self-consistent axis voting that aggregates local Gaussian predictions using confidence, offset proximity, and direction consistency.
- Achieved 2.43° mean angular error and 0.864 movable IoU on the PartNet-Mobility-derived benchmark; in a contextual comparison, symmetric axis distance was 0.063 versus 0.280 for 5-state ScrewSplat and 0.569 for SplArt.

Token-World

HMI Lab, School of Computer Science, Peking University

Compact VLM Tokens | Vision-Language-Action Models

Supervisor: Prof. Shanghang Zhang

- Applied an x_0 -parameterized flow-matching objective to model semantic token dynamics, and designed a GRU-based recurrent state to preserve past context across long-horizon rollout windows.
- Under open-loop action replay, outperformed WorldGym in VLM-feature fidelity (cosine 0.771 vs. 0.715; NMSE 0.489 vs. 0.616), reduced simulator step time to 0.359 s, and improved closed-loop evaluation correlation to $r = 0.794$ vs. 0.583 for Ctrl-World; submitted to ICRA 2027.

VA-CLIP

HMI Lab, School of Computer Science, Peking University

Multimodal Contrastive Learning | Representation Learning

Supervisor: Prof. Shanghang Zhang

- Used counterfactual action-video pairs to learn whether a predicted visual transition corresponds to the commanded robot action, then integrated the compatibility score into test-time rollout selection.
- Achieved over 86.3% R@5 retrieval accuracy, 96.7% counterfactual discrimination accuracy. Improved downstream WAM policy success

rate by 13% on eight contact-rich simulation tasks.

INDUSTRY EXPERIENCE

Muka Robotics
Research Intern

Jul. 2026 – Sep. 2026
Beijing

- Developing representation learning techniques and data cleaning pipeline for foundational world-action-model pretraining.
- Built a physical world model demo in which a robot observes, predicts, and plays tabletop curling against a human. [Video].

XtalPi

Research Intern, Future Chemistry Department

Mar. 2026 – Apr. 2026
Shenzhen

- Trained and deployed VLA policies, including LingBot-VLA, π_0 , and $\pi_{0.5}$, for robot manipulation in laboratory-automation settings.
- Integrated perception, policy inference, and control on real robot platforms; conducted iterative evaluation and ablations to diagnose deployment failures.

Beijing Academy of Artificial Intelligence

Research Intern, Embodied Multimodal Large Models Research Center

Jun. 2025 – Sep. 2025
Beijing

- Processed large-scale embodied datasets, including AgiBot-World and DROID, for VLA pretraining; implemented action normalization, tokenization, and structured training-data conversion.
- Reproduced representative VLA reinforcement-learning and robot-policy post-training frameworks; contributed to **FORCE** and **RoboBrain-X0** projects.

SELECTED PROJECTS

Awesome Representation for World Models
Open-Source Literature Curation

Aug. 2026 – Present
Project Lead

- Currently lead a collaborative survey of representations for world models, coordinating lab members to curate literature around state and transition representations.
- Designed a Cloudflare-based webpage workflow spanning literature review, taxonomy-based classification, and pull-request review for multi-contributor maintenance.

RoboBrain-X0: Cross-Embodiment Vision-Language-Action Model
Open-Source VLA Model

Jun. 2025 – Sep. 2025
Contributor

- Contributed to **RoboBrain-X0**, a unified VLA model for cross-embodiment action generation that supports heterogeneous robot platforms through a shared action-token representation.
- Focused on the data pipeline, processing AgiBot-World and DROID with action normalization, tokenization, and structured conversion for large-scale VLA pretraining.

LEADERSHIP AND SERVICE

Class Leader, Class 2303
Class Committee

2024 – 2025

- Led class operations and activities, including event planning, logistics coordination, routine administration, and award application materials; the class received honors including *Excellent Undergraduate Academic Style*, *Best Sports Team*, and *Best Popularity Class*.

House Division Lead, Street Dance Club
Student Organization

2024 – 2025

- Led the House division by coordinating weekly training with instructors and organizing two campus street dance performances, including rehearsal planning and member coordination.

HONORS AND AWARDS

- First Prize, Outstanding Student Award, SUSTech (top 5%; RMB 5,000 scholarship), Sep. 2025
- Third Prize, Outstanding Student Award, SUSTech (top 25%; RMB 1,000 scholarship), Sep. 2024
- First-Class Admission Scholarship, SUSTech (top 10%; RMB 50,000 scholarship), Sep. 2023
- Gold Medal, Chinese Chemistry Olympiad, Chinese Chemical Society (top 150 nationwide), Nov. 2022

SKILLS

- **Programming:** Python, PyTorch, Linux, Git
- **Robotics:** ROS, real-robot VLA deployment, camera calibration, 3D vision, and point-cloud processing
- **Languages:** Chinese (native); English